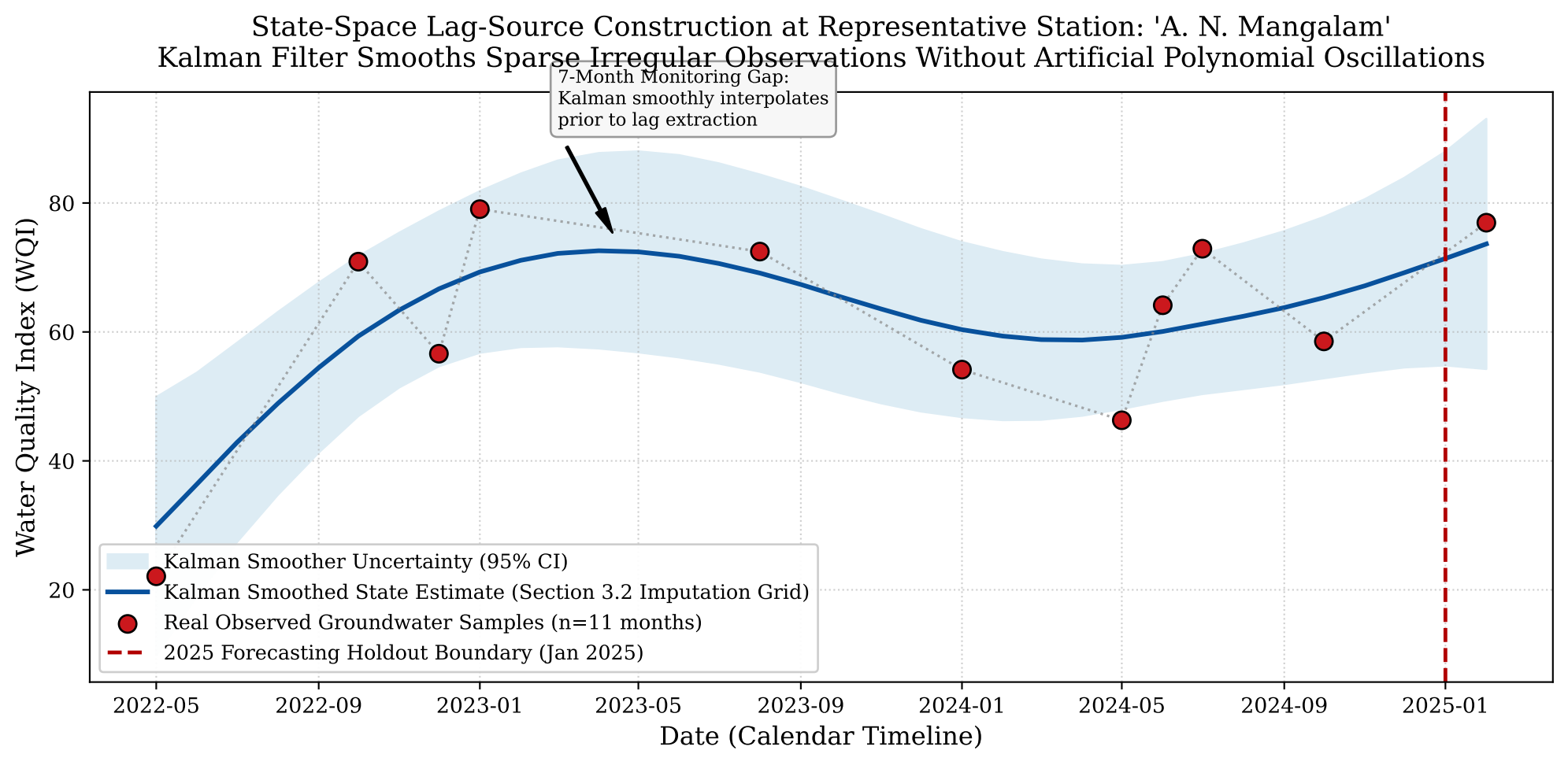
State-Space Lag-Source Construction at Representative Station: 'A. N. Mangalam'
Kalman Filter Smooths Sparse Irregular Observations Without Artificial Polynomial Oscillations
7-Month Monitoring Gap:
Kalman smoothly interpolates
prior to lag extraction
Water Quality Index (WQI)
Date (Calendar Timeline)
20
40
60
80
2022-05
2022-09
2023-01
2023-05
2023-09
2024-01
2024-05
2024-09
2025-01
Kalman Smoother Uncertainty (95% CI)
Kalman Smoothed State Estimate (Section 3.2 Imputation Grid)
Real Observed Groundwater Samples (n=11 months)
2025 Forecasting Holdout Boundary (Jan 2025)